12 / ROUTING

Department Routing Rules

How LangGraph selects the right expert agent

Purpose for the AI Assistant
The router node's rulebook. It reads the intent and entities from a question and sends it to exactly one department agent (or a small ordered set for cross-functional questions), avoiding the cost of consulting every agent.

1. Routing pipeline

```
Router flow
user question
-> normalize + synonym expand (13_SYNONYM_MAPPING.json)
-> intent + entity extraction (item? supplier? branch? metric?)
-> department scoring (keyword + term + entity signals)
-> pick agent (or ordered set for cross-functional)
-> agent runs its tools (16_LANGGRAPH_TOOL_DEFINITIONS)
-> compose 3-4 line answer
```

Single-agent by default
Most questions resolve to one agent. The router only fans out to multiple agents when the question is genuinely cross-functional (Section 4). Fanning out has a token and latency cost, so it is the exception, not the default.

2. The six department agents

Inventory (Stores) Agent

Trigger keywords	stock, available, hold, reorder, warehouse, on hand, blocked, inventory value, critical
Business terms	Available Stock, Hold Stock, Inventory Value, Reorder Level, Stock Health, Critical Item
Tables it owns	stock, ab_items
Decision rules	Route here when the question is about what we physically have, what it is worth, or whether it is running low. If the metric is Stock Health / Critical / Safety Stock, use ab_items; otherwise use stock.

Purchase Agent

Trigger keywords	supplier, vendor, PO, purchase order, buy, procurement, delay, lead time, spend, price
Business terms	Supplier Delay, Purchase Value, Lead Time, MOP, Supplier Ranking
Tables it owns	purchase

Decision rules	Route here for buying activity and supplier performance. “Late supplier” / “delayed vendor” → supplier delay logic. Local procurement only — for overseas suppliers see the Import Agent.

Import Agent

Trigger keywords	import, LC, ALC, on water, sailing, ETA, ETD, shipment coming, customs, China, foreign supplier
Business terms	On Water, Sailing (inbound), ETA/ETD, LC/ALC, ETA Works, Country Dependency
Tables it owns	import_shipment
Decision rules	Route here when goods are INBOUND from a foreign supplier. “Sailing” + a supplier/PO/country context → Imports. Payment-instrument words (LC/ALC) are always Imports.

Logistics Agent

Trigger keywords	export, customer shipment, container, BL, GD, POD, freight, transporter, truck, dispatch, sailing date
Business terms	Shipment Delay, Freight/kg, Documentation Completion, Sailing (export), Transit Time
Tables it owns	logistics_shipment
Decision rules	Route here when goods are OUTBOUND to a customer, or for internal shifting between branches. “Sailing” + a customer/export/vessel-departure context → Logistics.

Issuance (Consumption) Agent

Trigger keywords	consumption, usage, issued, consumed, department used, machine used, job cost, burn
Business terms	Consumption Value (TotalPric), Department/Machine/Job Consumption
Tables it owns	issuance
Decision rules	Route here for anything about material leaving stores for use. Value questions use SUM(total_pric).

Item Master Agent

Trigger keywords	item code, specification, category, what is item, classify, unit, group
Business terms	Item Code, Specification, Item Sub Group, Material Standard
Tables it owns	item_master
Decision rules	Route here for identity/classification questions, and as the entity-resolution helper other agents call before running SQL.

3. The ‘sailing’ disambiguation (worked example)

“How many items are sailing?” can mean Imports or Logistics. The router resolves it by context, not by the word alone.

```
Decision rule
if context mentions supplier / PO / LC / country / 'coming' / 'arriving':
-> IMPORT AGENT (inbound shipment on water)
elif context mentions customer / export / vessel / dispatch / BL:
-> LOGISTICS AGENT (outbound export vessel departure)
else:
-> ASK ONE CLARIFYING QUESTION:
'Do you mean incoming imports or outgoing export shipments?'
```

Term meaning is fixed in the dictionary
“On water” always = import In-Transit. “Sailing” is the only truly ambiguous term and is the one case where the router may ask before answering.

4. Cross-functional routing

Some management questions need more than one agent. The router runs them in a fixed order and the master agent composes one answer.

Question	Agents (in order)	Why
Which materials may go out of stock?	Inventory → Import	on-hand + criticality, then check incoming shipments
Which suppliers create production risk?	Purchase → Import → Inventory	delay + country risk + which critical items they supply
Which items to prioritise for procurement?	Inventory → Issuance → Import	criticality + burn rate + what's already in transit
Why did inventory value increase?	Purchase → Issuance	value in (purchases) vs value out (issues)
Full picture for item X	Item → Inventory → Issuance → Purchase → Import	identity, then every domain keyed by item_code

5. Scoring & tie-breaks

- Score each agent = (keyword hits × 1) + (business-term hits × 2) + (resolved-entity match × 3).
- Highest score wins; a margin below a threshold triggers a clarifying question.
- Entity type is the strongest signal: a resolved supplier → Purchase/Import; a resolved item → Inventory/Item.
- LC / ALC / on-water → always Imports. BL / GD / POD / freight → always Logistics.
- If the question is a pure Definition (‘what is X’), skip agents — answer from the Business Dictionary.

6. Router output contract

```
Structured router decision (passed to the master agent)
{
"intent_class": "aggregate|lookup|ranking|derived_metric|diagnostic|definition",
"primary_agent": "inventory|purchase|import|logistics|issuance|item",
"secondary_agents": ["import"], // ordered, only if cross-functional
"entities": {"item_code": "19981-60", "branch": "QCL"},
"needs_clarification": false,
"clarifying_question": null
}
```